STAR TOPOLOGY

A **Star Topology** is a network configuration in which all devices are connected individually to a **central networking device**, such as a **switch** or **hub**. The central device acts as the main point of communication and manages the transmission of data between the connected devices. Each device has a separate connection to the central device, making the network easy to manage and expand.
In a star topology, devices communicate with each other through the central switch or hub. If one device or its connecting cable fails, the other devices in the network can continue to communicate normally. However, if the central device fails, communication between all connected devices is disrupted. This makes the central device a critical component of the network.
Star topology is widely used in modern **Ethernet** networks because it provides better reliability, easier troubleshooting, and flexible network management. It is commonly used in schools, offices, universities, computer laboratories, and homes where multiple computers and other network devices need to be connected through a central switch or hub.

Advantages of Star Topology

- Provides better network reliability because the failure of one device or its connecting cable does not affect the other devices connected to the network.
- Makes network management easier because all devices are connected through a central switch or hub, allowing the network to be monitored and maintained efficiently.
- Simplifies troubleshooting because each device has a separate connection to the central device, making it easier to identify and isolate connection problems.
- Provides better network performance because devices have dedicated connections to the central switch, reducing collisions and improving data transmission.
- Easy to expand because new devices can be added to the network by connecting them directly to the central switch or hub without affecting the existing connections.
- Provides flexibility by allowing different types of devices to be connected and managed through a central networking device according to the requirements of the network.

Disadvantages of Star Topology

- More expensive to set up because star topology requires a central networking device, such as a switch or hub, along with separate cables for each connected device.
- Requires additional cabling because every device must have an individual connection to the central switch or hub.
- The central networking device is a single point of failure, so if the switch or hub fails, communication between all connected devices is disrupted.
- Network performance depends on the capacity of the central device, and an overloaded switch or hub can affect the overall network performance.
- Requires proper configuration and management of the central networking device to ensure reliable communication between connected devices.
- As the number of devices increases, more ports and cables are required, which can increase the cost and complexity of the network.
- A failure in the cable connecting an individual device affects that device's connection to the network, although the remaining devices continue to operate normally.

Simulation / Design in Cisco Packet Tracer

This section presents the **Star Topology** network designed and simulated in Cisco Packet Tracer. The network consists of multiple **end devices** connected individually to a **central switch**, which acts as the main communication point for the connected devices. Each device communicates with other devices through the central switch, representing the basic structure and working of a Star Topology. This arrangement demonstrates how devices can be connected, managed, and communicated within a network using a centralized networking device.

Network Diagram

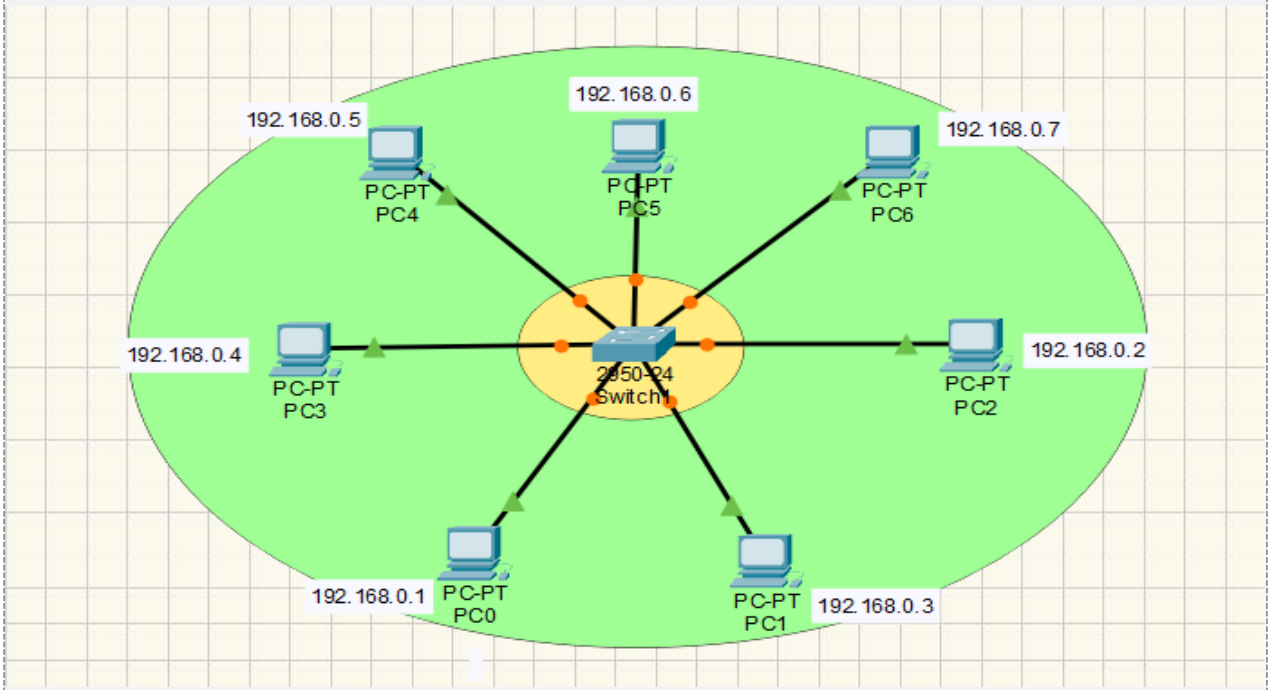


Figure 1: Star topology layout in Cisco Packet Tracer

Device Configuration

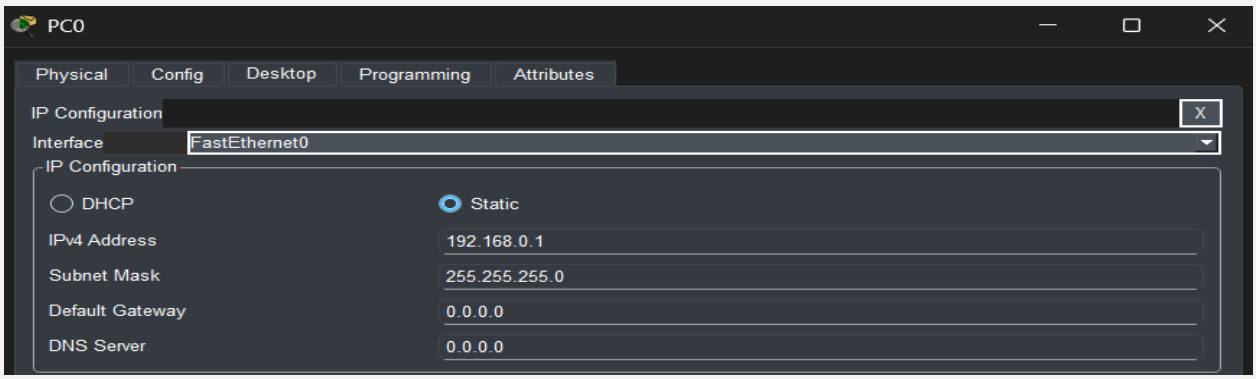


Figure 2.1: IP addressing / device configuration window

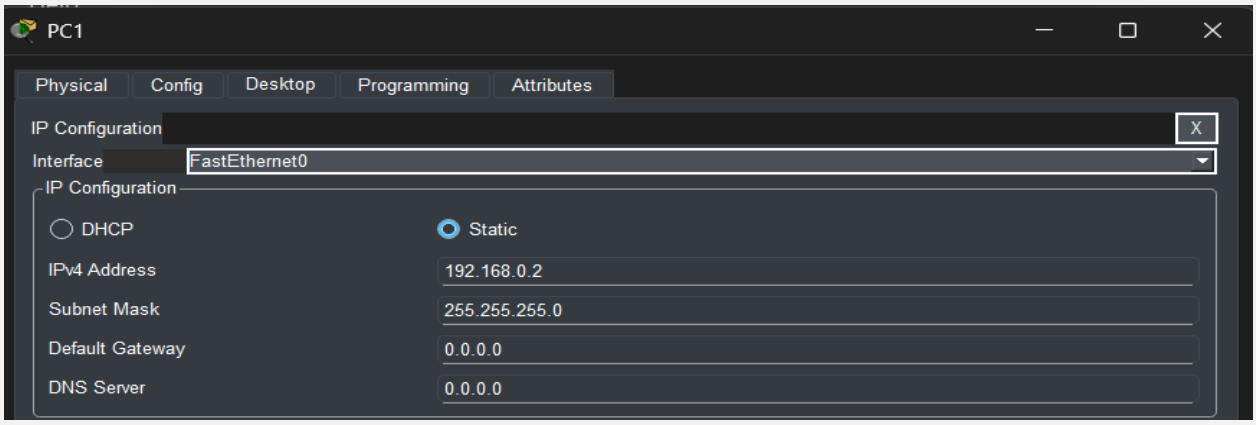


Figure 2.2: IP addressing / device configuration window

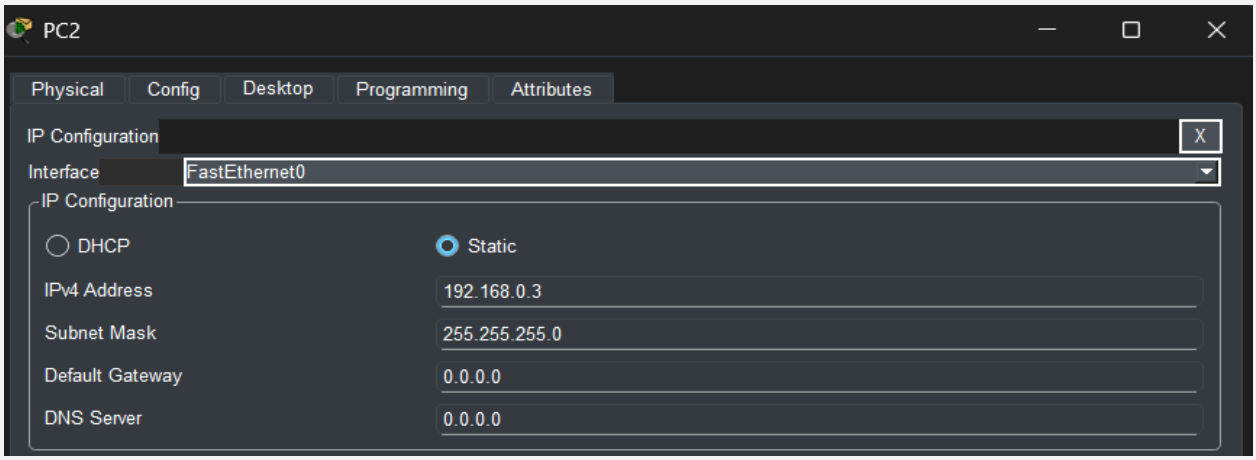


Figure 2.3: IP addressing / device configuration window

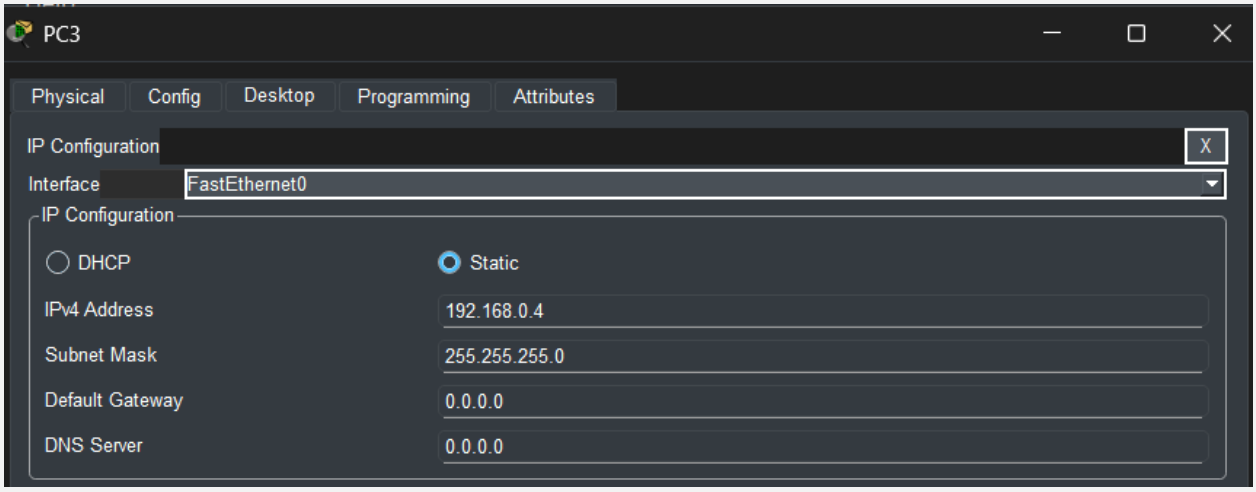


Figure 2.4: IP addressing / device configuration window

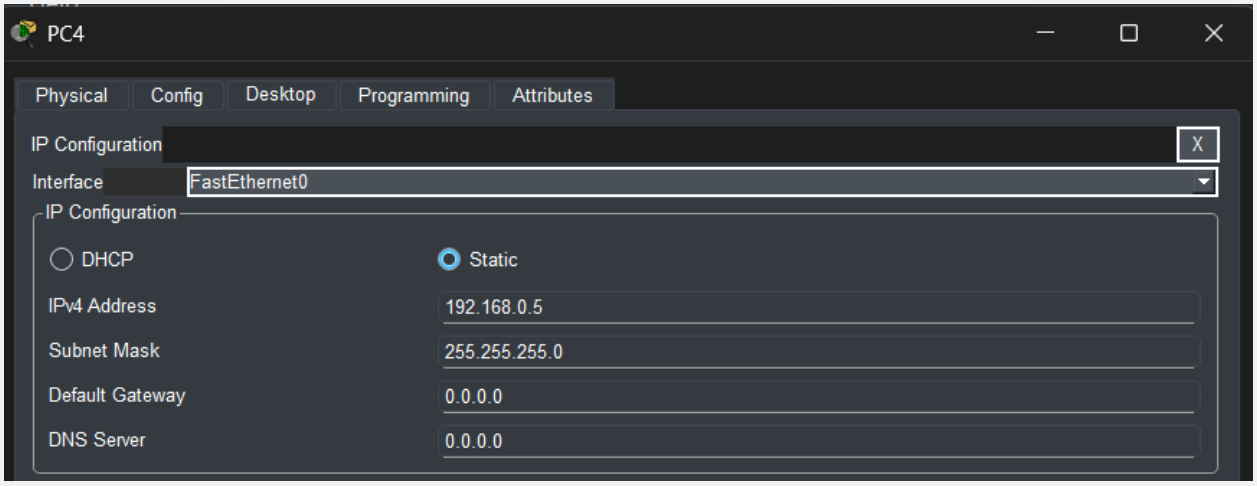


Figure 2.5: IP addressing / device configuration window

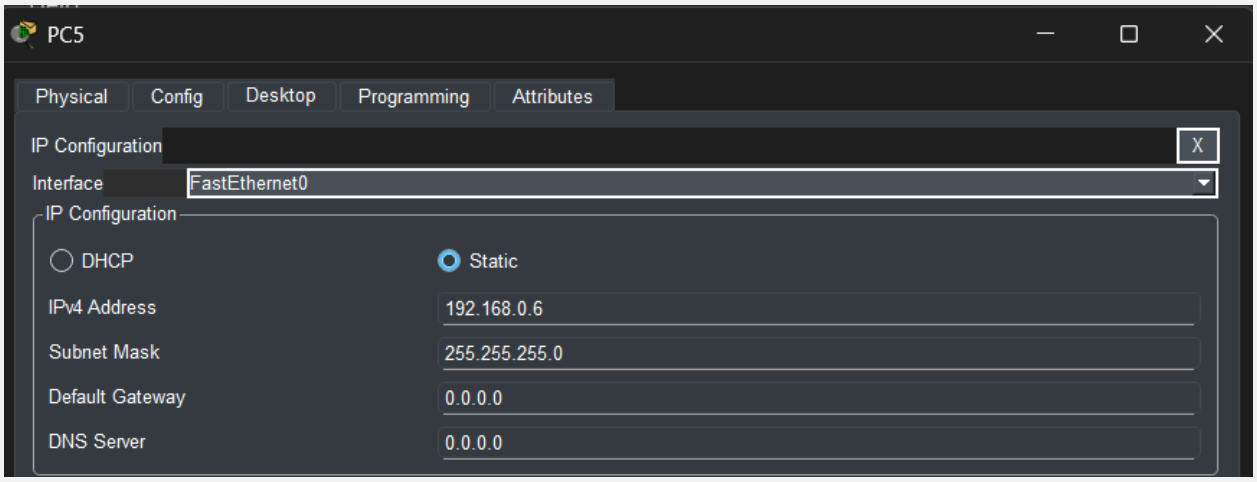


Figure 2.6: IP addressing / device configuration window

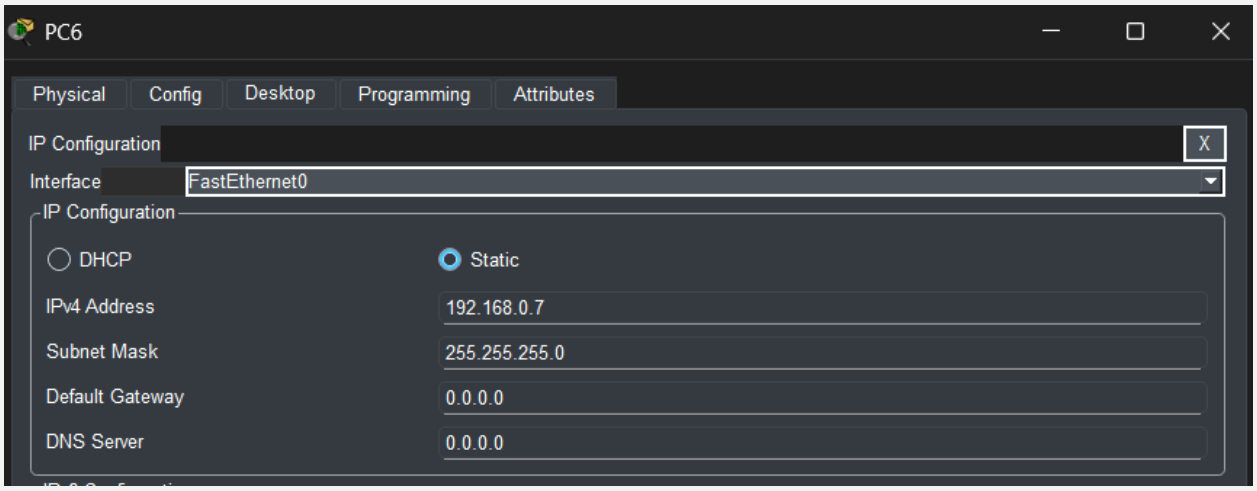


Figure 2.7: IP addressing / device configuration window

Connectivity Test (Ping / Simulation Result)

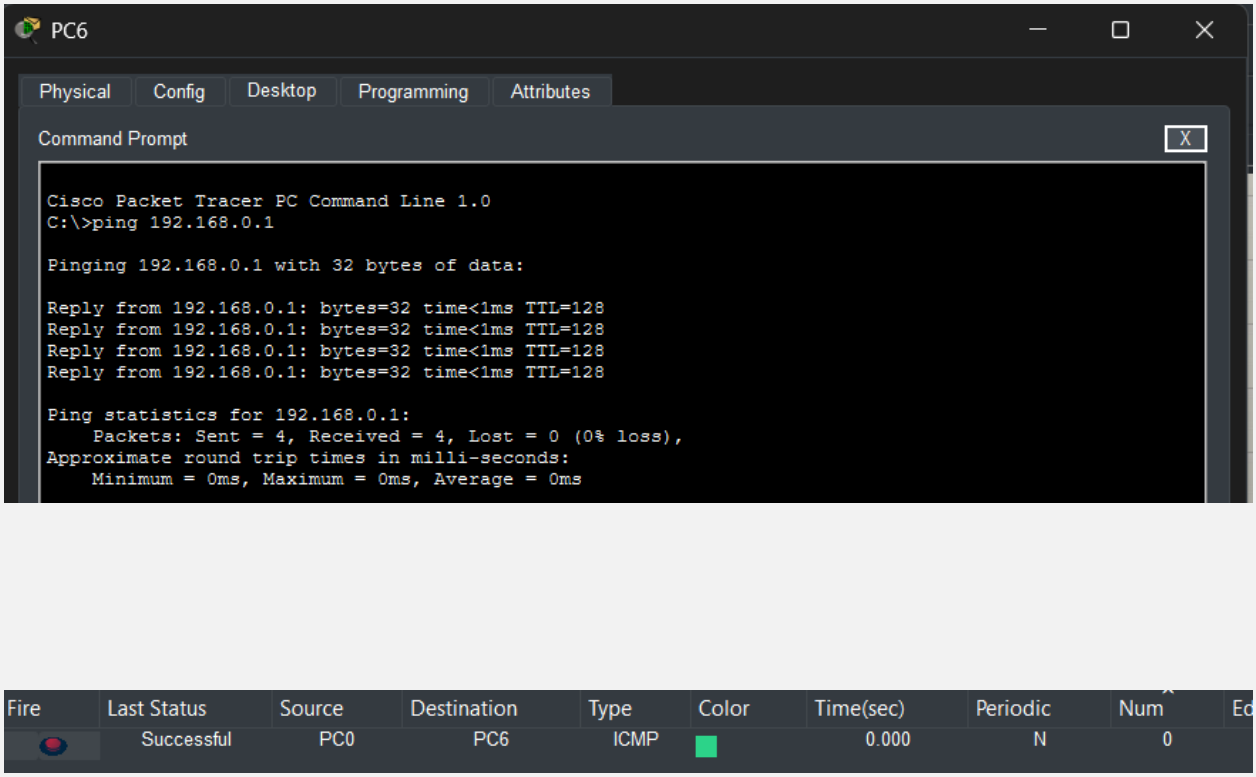


Figure 3: Successful ping / packet simulation result between PCs

Process of Designing the Network in Packet Tracer

- Open Cisco Packet Tracer and create a new blank project to begin designing the Star Topology.
- Select a suitable switch from the Network Devices category and place it at the center of the workspace. The switch acts as the central networking device of the Star Topology.
- Select the required end devices, such as PCs, from the End Devices category and place seven PCs on the workspace. The PCs are arranged around the central switch to give the network the appearance of a Star Topology.
- Select Copper Straight-Through cable from the Connections category and use it to connect each PC individually to the central switch. Each PC must have a separate physical connection to the switch.
- Connect PC0 to PC6 to separate FastEthernet ports on the switch. The completed physical topology should consist of seven PCs connected directly to one central switch.
- Configure the network settings of each PC by opening Desktop → IP Configuration. A unique IP address and a suitable subnet mask are assigned to every PC, with all devices placed within the same network so that they can communicate with one another.
- The IP addresses are assigned as follows: PC0 is configured with 192.168.0.1, PC1 with 192.168.0.2, PC2 with 192.168.0.3, PC3 with 192.168.0.4, PC4 with 192.168.0.5, PC5 with 192.168.0.6, and PC6 with 192.168.0.7. The subnet mask 255.255.255.0 is used for all PCs.
- After completing the connections and configurations, check the link indicators between the PCs and the switch. Green indicators show that the physical connections are active and the devices are properly connected.
- The network can then be tested using Simulation Mode. The Add Simple PDU option is used to send a test packet from one PC to another, allowing the movement of the packet through the central switch to be observed.
- Connectivity can also be checked through the Command Prompt of a PC by using the ping command with the IP address of another PC. For example, from PC0, a ping can be sent to PC6 using the command ping 192.168.0.7. A successful reply confirms that communication between the two PCs has been established.
- Additional connectivity tests can be performed between different PCs to verify that all devices connected to the central switch can communicate with one another.
- Screenshots of the completed Star Topology, IP configurations, and successful connectivity tests are captured as evidence of the completed network setup.
- Finally, the completed Packet Tracer project is saved as a .pkt file, along with the required documentation and screenshots for submission.

Conclusion

The Star Topology was successfully designed and configured using Cisco Packet Tracer, demonstrating how multiple devices can be connected through a central networking device. The experiment involved connecting seven PCs to a central switch, configuring IP addresses, and establishing individual connections between each PC and the switch. Connectivity was then verified using the ping command and Simulation Mode. This experiment helped reinforce the practical understanding of Star Topology, centralized network communication, device connectivity, and basic network configuration.